Innovative Evaluation Practices for

BCSE308P Computer Networks Lab

Fall Semester 2026-27

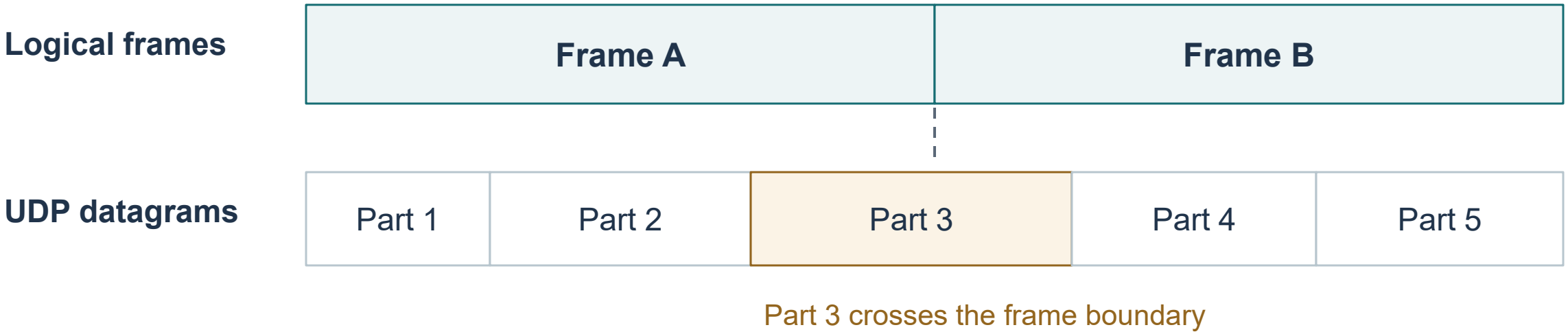
CCSDS INSPIRED SOFTWARE TELEMETRY RECEIVER AND GROUND STATION

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Submitted to: SREETHAR S.

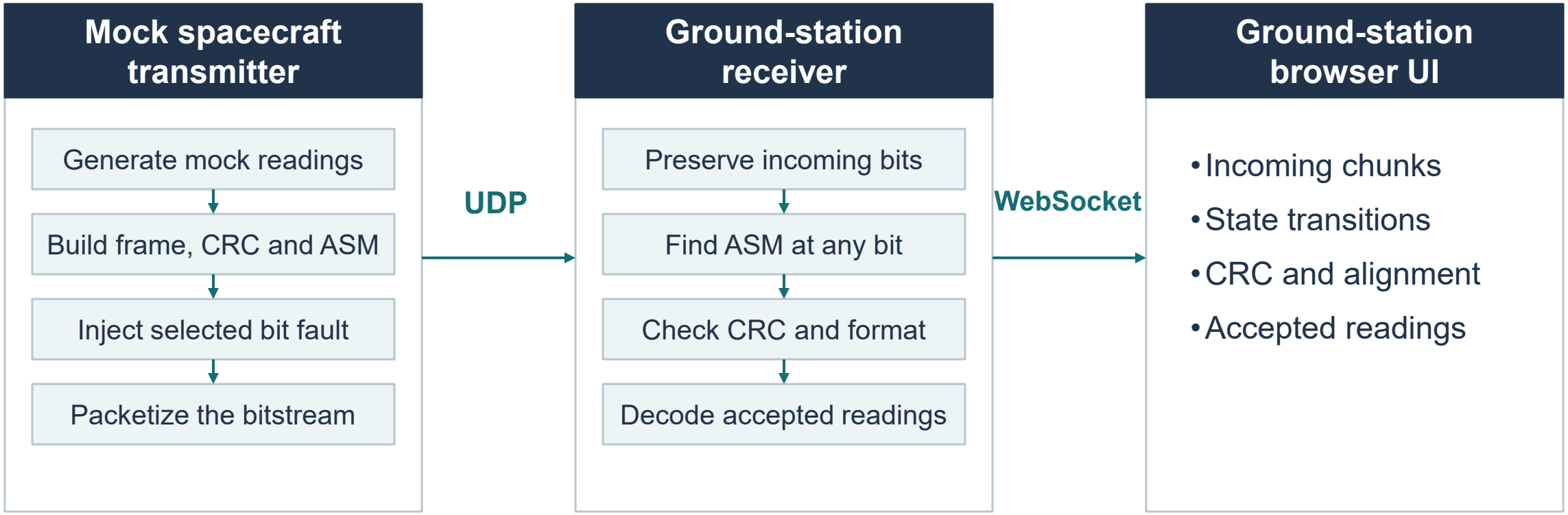
Frame boundaries and arriving messages

- A telemetry frame can span several arriving messages.
- One message can contain the end of one frame and the start of another.
- The receiver must recover boundaries after corruption or a bit slip.



- Our packetizer creates the chunks. UDP carries each chunk as one datagram.

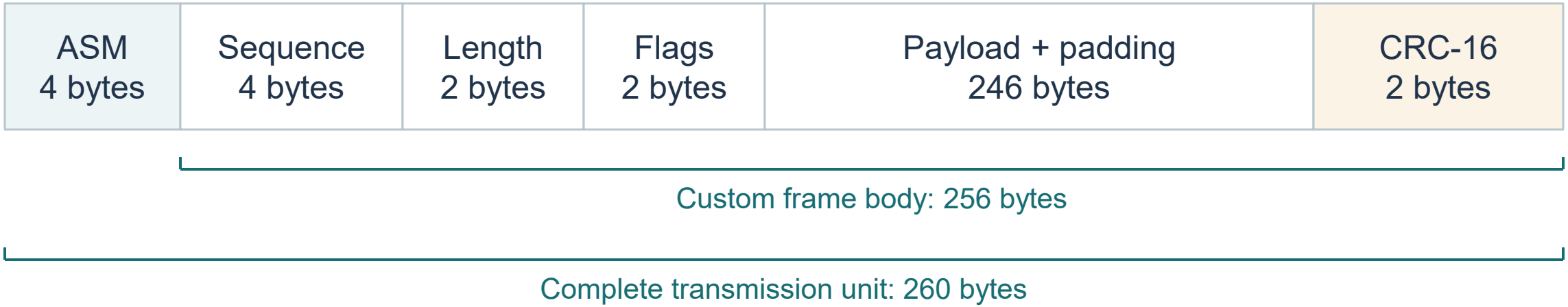
System architecture



- UDP carries telemetry chunks. WebSocket carries receiver observations.

Telemetry transmission unit

260 bytes = 2,080 bits



- ASM marks a possible start. Sequence, length and flags form the 8-byte header.
- The default payload contains 12 telemetry bytes and 234 zero-padding bytes.
- The CRC is an additional check value. Readings remain in the payload.

Persistent buffering between arrivals

- The buffer retains unfinished data between datagrams.
- New bits are appended. Processed bits are consumed or discarded.
- A complete candidate needs 2,080 bits, starting at its ASM.



1,752 retained bits + 384 new bits = 2,136 available bits



Hypothetical clean candidate: 56 bits remain after a valid unit is consumed

```
00000000 00000001 00000000 00001100 00000000 00000000 00001001 01100101
```

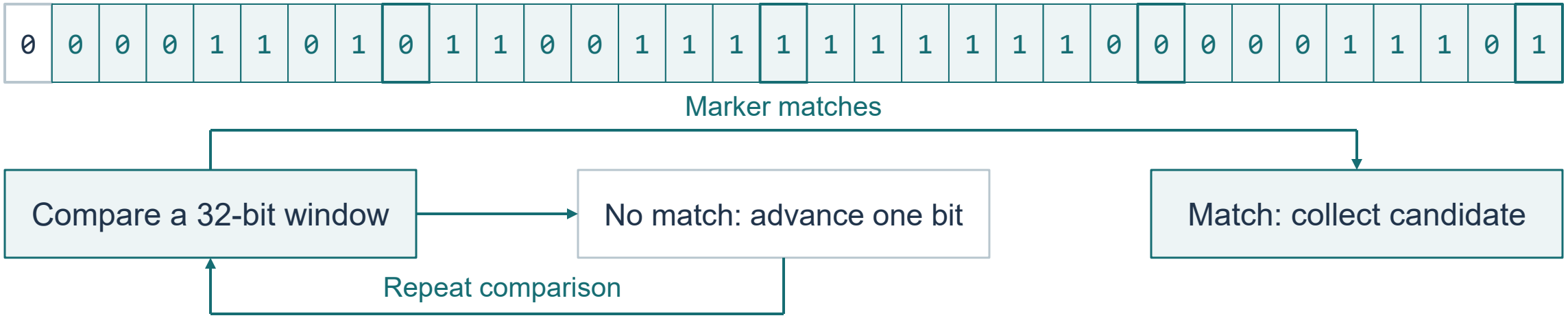
Attached Sync Marker search

`ASM is a fixed 32-bit marker: 0x1ACFFC1D`

Exact marker bits

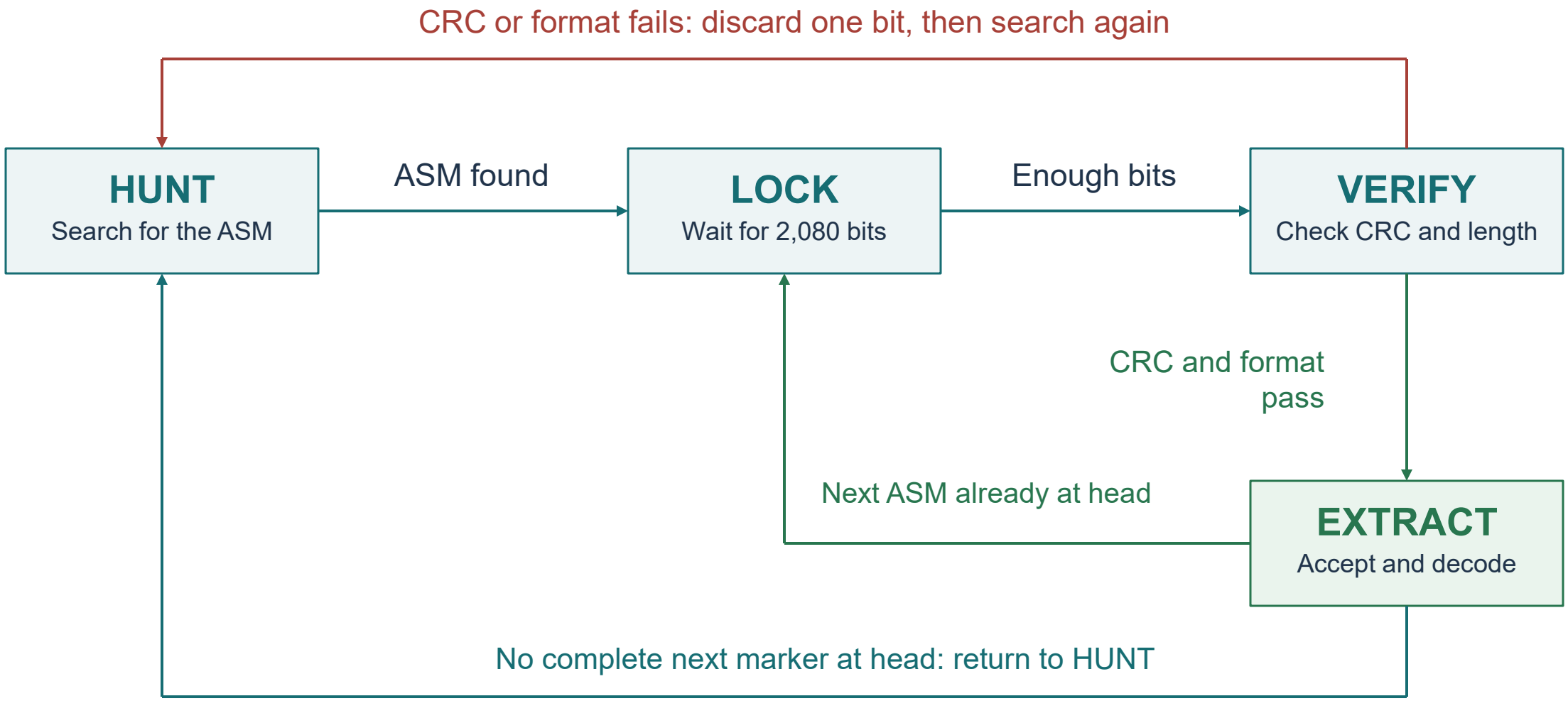


Same marker beginning one bit later in a schematic stream



The marker can cross byte or datagram boundaries. CRC and format checks still follow.

Receiver finite-state machine



Marker alignment inside a byte

Alignment is the marker start position modulo 8

Byte boundary: alignment 0



Marker starts at the first bit

Last bit of a byte: alignment 7

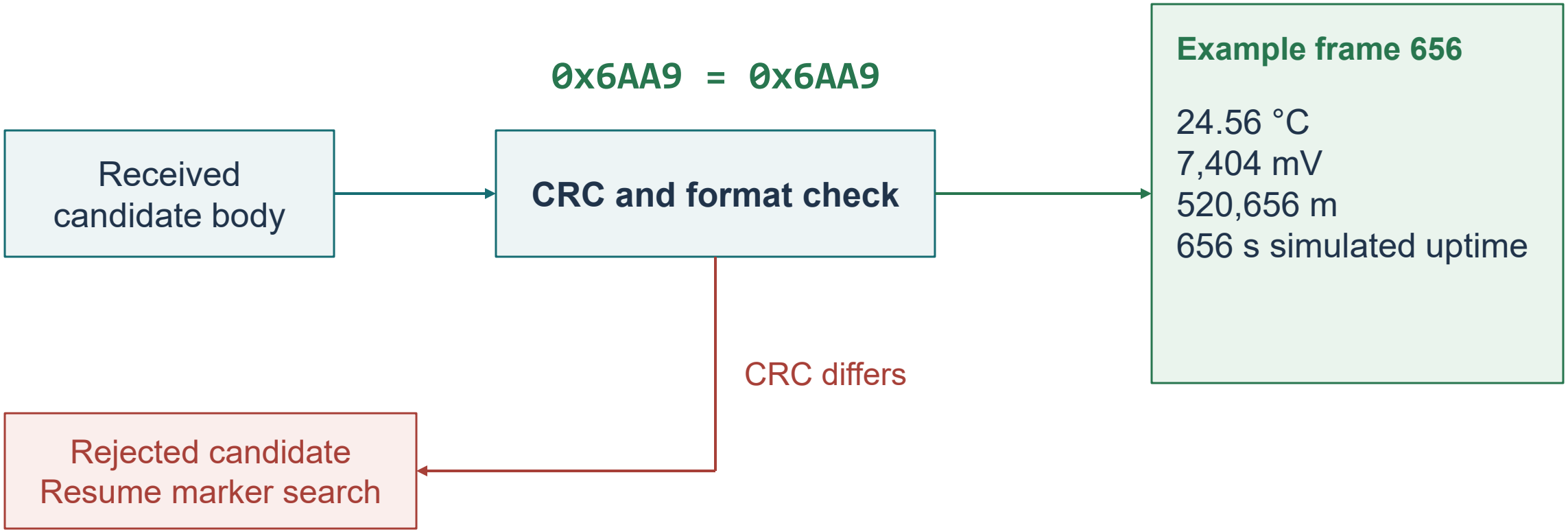


Marker continues into later bytes

- Stream bit 4,160 has alignment 0 because 4,160 mod 8 = 0.
- Stream bit 8,319 has alignment 7 because 8,319 mod 8 = 7.
- Alignment does not identify a frame number or the damaged bit.

CRC validation and decoded telemetry

CRC = Cyclic Redundancy Check



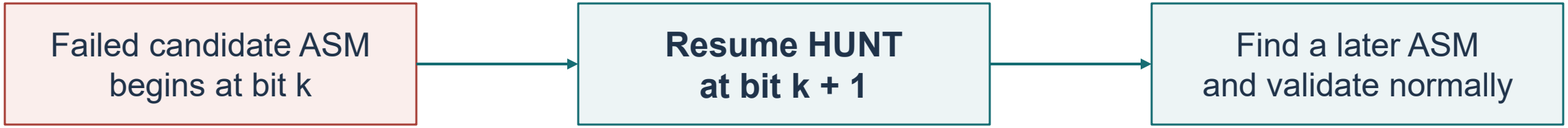
- Only accepted frames update the displayed readings.
- A matching CRC means no corruption was detected. It does not repair data.

Recovery after a damaged candidate

The receiver rejects the candidate and keeps running



Bit-by-bit search through the remaining buffered data



- Search restarts one bit after the failed marker, not 2,080 bits later.
- Recovery skips damaged readings. It does not reconstruct them.

Effects of the guided experiments

Experiment	Change in the logical stream	Later alignment
Clean	No fault, with normal chunk boundaries	0
Start between bytes	Prefix three bits before the stream	3
Flip one bit	Change one value without changing bit count	0 then 0
Insert one bit	Add one bit and move later starts forward	0 then 1
Delete one bit	Remove one bit and move later starts backward	0 then 7

Schematic edits to original bits 1 0 1 1 0 0

Flip

```
1 0 [0] 1 0 0
```

Insert

```
1 0 [1] 1 1 0 0
```

Delete

```
1 0 [ ] 1 0 0
```

Six-frame clean / offset runs: 6 accepted, 0 CRC failures

Flip / insert / delete: 5 accepted, 1 CRC failure, 1 gap, 1 reacquisition

Demonstration workflow

1 Clean transmission

Buffer fills and a frame passes

2 Flip one bit

CRC mismatch rejects frame 2

3 Insert one bit

Alignment 1 and valid frames resume

Controls: Restart, Play/Pause, Step, Next state, and playback speed

- Guided lab processes a six-frame recording in memory through the same receiver engine.
- Live mode receives actual localhost UDP datagrams and publishes observations over WebSocket.

Terminal 1

```
npm run receiver
```

Terminal 2

```
npm run ui
```

Browser

```
http://127.0.0.1:5173
```

```
Optional Terminal 3 for Live: npm run transmitter -- --mode clean
```

The receiver continues decoding later valid frames without restarting